
Kuber Kumar Chaurasiya

Lead AI & ML Engineer | Generative AI, Semantic Search & Enterprise AI Systems

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EXECUTIVE SUMMARY

Lead AI/ML Engineer with 14 years of experience building large-scale machine learning, NLP, search and Generative AI solutions across healthcare, advertising, and operations. Experienced in developing production-grade RAG systems, semantic search platforms, and LLM-powered applications. Proven track record of delivering scalable AI systems with measurable business impact through strong expertise in AI architecture, retrieval systems, and applied machine learning.

CORE COMPETENCIES

- **AI & Machine Learning:** Generative AI | RAG Systems | Semantic Search | NLP | Deep Learning | Recommendation Systems | LLM Fine-tuning | Information Retrieval
- **AI Engineering & Architecture:** LangChain | LangGraph | AI Agents | Vector Databases | Prompt Engineering | Retrieval Optimization | AI System Design | Model Evaluation | AI-assisted development
- **Engineering & Cloud:** Python | FastAPI | PyTorch | AWS (SageMaker, Bedrock, EMR) | Google Cloud | Docker | Kubernetes | API Development | MLOps
- **Leadership & Collaboration:** Technical Leadership | AI System Architecture | Cross-Functional Collaboration | Team Mentoring

PROFESSIONAL EXPERIENCE

Carelon Global Solutions | AI Tech Lead | Bengaluru, IN | Aug 2021 – Present

Led the development of RAG-based AI assistants and semantic search systems for healthcare support, enabling faster query resolution, improved search relevance, and scalable LLM-powered customer support workflows.

Claims Support AI Assistant

- Built and deployed an **AI assistant** using **LangGraph, OpenAI models, and vector search** to help customer care teams answer claims and policy-related questions.
- Developed document processing and retrieval pipelines for healthcare knowledge sources including policy documents and SOPs.
- Implemented **hybrid retrieval and semantic search** using dense embeddings and metadata-aware retrieval strategies to **improve answer grounding** and retrieval relevance.
- **Created test cases** from historical user queries and **built automated evaluation pipelines using RAGAs** to compare LLMs, embeddings, and retrieval approaches.
- Developed **scalable inference APIs** and enterprise AI workflows for real-time query handling and response generation using FastAPI.
- Reduced average call handling time by 22% and contributed to an estimated annual operational savings of approximately \$200K.

Semantic Search & Provider Discovery Platform

- Led a team of 5 data scientists to modernize **healthcare provider search using transformer-based semantic retrieval** and embedding models.
- Built dense retrieval pipelines using Sentence Transformers models including all-mpnet-base-v2 and KRISSBERT to improve mapping between user queries, medical specialties.
- Developed fine-tuned embedding models using SentenceTransformers on internal healthcare datasets to improve semantic relevance and reduce irrelevant search results.
- Built a **Streamlit-based SME evaluation platform** to capture relevance feedback and create evaluation datasets for measuring retrieval quality and model performance.
- **Improved search relevance metrics** including NDCG by 15%, increasing successful searches and enhancing member search experience.

Accenture | Data Science Associate Principal | Bengaluru, IN | Nov 2017- Aug 2021

Built large-scale AI solutions for finance and accounts payable automation

- Developed a **duplicate invoice detection solution** using **deep learning, fuzzy matching**, and similarity-based ranking techniques to **capture 96% of duplicate payments** within the top 50% highest-risk invoices, significantly reducing manual audit effort and cost.
- Built a large scale **multiclass classification model** on 100Mn+ invoices to predict categories of each invoice using Python, vowpal wabbit and xgboost.

Vserv Digital Services | Data Scientist | Mumbai, IN | Mar 2015 - Nov 2017

- Developed large-scale **CTR prediction models** using **Python and Hive in AWS EC2/EMR** resulting in 8% increase in clicks and 20% boost in conversions.

Additional early-career analytics experience (2011–2015) across TCS, OnMobile, and Wize Commerce, applying statistical modeling, predictive analytics and churn analysis.

EDUCATION, CERTIFICATIONS

- **M.Sc. (Applied Statistics & Informatics)** | IIT Bombay | 2011
- **B.Sc. (Statistics Hons.)** | BHU | 2009
- **Certifications:** AWS Certified [AI Practitioner](#), Agentic System Design([Educative](#)), NLP Nanodegree([Udacity](#)), Deep Learning Specialization([Coursera](#))
- **Kaggle Notebooks, Discussion expert** | 2021

EXTRACURRICULAR ACTIVITIES

- **Held multiple leadership roles**(VP Education, VP Membership) at **Leaders Speak Toastmasters Club** | 2024-2025
- Reading Books(self-help), Trekking, Personal Finance, Mentoring juniors